Dear TA Quant Team,

I want to work with you.

I read that you have built one of the greatest teams of developers and data scientists in HFT. You have a machine learning expert and top-level data scientist as your founder. You trade across traditional and crypto markets. You have cutting-edge hardware. You give 40 paid days off.

You are exactly where I belong.

Here is who I am:

I am from Tunisia. I dropped out of mechanical engineering because I fell in love with trading. Not the money. The game. The analysis. The puzzle.

I have built trading systems alone. From scratch. Every part.

I have built statistical arbitrage bots. Trend following bots. Mean reversion bots. Every kind of bot you can imagine.

My latest system:
- Systematic strategy with 3-factor model
- Full automation from signal to execution
- Python backend with Flask
- 4,320 parameter combinations tested

I am calm under pressure. I am analytical. I am a perfectionist. When I love something, I give it everything.

I know you ask for 3 years of professional experience. I don't have that. I have 3 years of building systems alone. In Tunisia. With no team. With no resources. That is not less experience. It is harder experience.

I want to work with people who understand me. People who think like me. People who live in the quant world.

Let me show you what I built.

THE STRATEGY LOGIC - 1:1 RISK-REWARD

I coded a Pine Script strategy that uses three confirmations to enter trades on a 1-minute timeframe:

1. Trend Direction: Price above or below the input-period EMA
2. Key Level: Price touches the Donchian Channel
3. Price Action Confirmation: Pin bar (body ≤33% of range, wick ≥2x body) OR engulfing pattern (close beyond previous high/low, open opposite)

When all three align, the strategy enters a trade. The exit follows a strict 1:1 risk-reward ratio:
- Stop loss = entry ± stop distance
- Take profit = entry ∓ stop distance
- Both are equal distance from entry

This means every trade risks the same amount it aims to gain. The 1:1 ratio simplifies risk management and makes the Martingale mathematics clean.

```
//@version=5
strategy("1-Min Scalper (ATR Stop + Bucket Martingale)", overlay=true, initial_capital=10000, defaul

// --- Inputs ---
ema_len        = input.int(50, "EMA Length")
donchian_len   = input.int(20, "Donchian Length (Key Level)")
use_pin        = input.bool(true, "Use Pin Bar Pattern")
use_engulf     = input.bool(true, "Use Engulfing Pattern")
atr_len        = input.int(14, "ATR Length")
atr_mult       = input.float(1.5, "ATR Multiplier for Stop")
point_value    = input.float(10, "Point Value ($ per 1 point/pip per contract)")

// --- Time Control (UTC) ---
utcHour   = hour(time, "UTC")
utcMinute = minute(time, "UTC")
close_time = (utcHour == 20 and utcMinute >= 50)
no_trade = (utcHour == 20 and utcMinute >= 50) or (utcHour == 21 and utcMinute < 5)

// --- Calculations ---
ema = ta.ema(close, ema_len)
donch_high = ta.highest(high, donchian_len)
donch_low  = ta.lowest(low, donchian_len)
atr = ta.atr(atr_len)

// --- Pattern Definitions ---
is_pin_bar(candle_high, candle_low, candle_close, candle_open) =>
    body = math.abs(candle_close - candle_open)
    upper_wick = candle_high - math.max(candle_close, candle_open)
    lower_wick = math.min(candle_close, candle_open) - candle_low
    body <= (candle_high - candle_low) * 0.33 and (lower_wick > body * 2 or upper_wick > body * 2)

is_bull_engulf() =>
    close > high[1] and open < close[1]

is_bear_engulf() =>
    close < low[1] and open > close[1]

// --- Entry Conditions ---
long_trend   = close > ema
```

RISK MANAGEMENT - MARTINGALE WITH 1:1 RR AND COMMISSION NORMALIZATION

First, understand the core idea:

I risk small amounts. If I lose, I increase the next trade. When I finally win, I recover all previous losses AND make a profit.

The Risk Progression:

Step 1: Risk 1 unit
Step 2: Risk 3 units
Step 3: Risk 7 units
Step 4: Risk 15 units
Step 5: Risk 31 units
Step 6: Risk 63 units
Step 7: Risk 127 units

The Rules:
- If I **lose** → next trade uses higher risk
- If I **win** → next trade resets to 1 unit
- If I **lose** 7 times in a row → reset to 1 unit and continue

The Mathematics:

With a 50% win rate, here is what happens in 300 trades per instrument per week:

- Most trades win immediately → profit **+1 unit**
- Some trades lose 1 then win → profit **+2 unit**
- Some lose 2 then win → profit **+3 unit**
- Some lose 3 then win → profit **+4 unit**
- Some lose 4 then win → profit **+5 unit**
- Some lose 5 then win → profit **+6 unit**
- Some lose 6 then win → profit **+7 unit**
- Very rarely, 7 losses in a row → loss of **-247 unit**

Because the win rate is **above 50%**, the profits from recovery sequences outweigh the rare 7-loss events.

What This Means Per Trade:

Every trade is designed to earn a minimum of 0.8 unit net after commission. Whether I win or lose, the structure ensures that when I eventually win, I earn more than I lost.

The Problem - Spread and Commission:

In normal accounts, spread and commission eat profits. I solved this by:

1. Choosing a raw spread account with near-zero spread and 6$ commission per lot.
2. Using a broker that allows nano lots (0.001 minimum)
3. Creating a table that builds commission into every calculation

The Solution - My Normalization Table:

I created a table that calculates exact lot sizes for every step and every possible stop distance. Commission is included. Net profit after commission is always at least $0.40 per winning sequence.

STEP 1 (0.5)						
Stop Range (ticks)	Lot Size	Capped Stop	Risk	Commission	Won	Lost
250+	0,002	250	0,500	0,012	0,488	0,512
166 - 250	0,003	166	0,500	0,018	0,482	0,518
125 - 166	0,004	125	0,500	0,024	0,476	0,524
100 - 125	0,005	100	0,500	0,030	0,470	0,530
83 - 100	0,006	83	0,498	0,036	0,462	0,534
71 - 83	0,007	71	0,497	0,042	0,455	0,539
62 - 71	0,008	62	0,496	0,048	0,448	0,544
55 - 62	0,009	55	0,495	0,054	0,441	0,549
50 - 55	0,010	50	0,500	0,060	0,440	0,560
45 - 50	0,011	45	0,495	0,066	0,429	0,561
41 - 45	0,012	41	0,492	0,072	0,420	0,564
38 - 41	0,013	38	0,494	0,078	0,416	0,572
38-	0,014	35	0,490	0,084	0,406	0,574

STEP 2 (1.7)						
250+	0,007	250	1,750	0,041	1,709	1,791
166 - 250	0,010	166	1,660	0,061	1,599	1,721
125 - 166	0,014	125	1,750	0,082	1,668	1,832
100 - 125	0,017	100	1,700	0,102	1,598	1,802
83 - 100	0,020	83	1,660	0,123	1,537	1,783
71 - 83	0,024	71	1,704	0,144	1,560	1,848
62 - 71	0,027	62	1,674	0,165	1,509	1,839
55 - 62	0,031	55	1,705	0,185	1,520	1,890
50 - 55	0,034	50	1,700	0,204	1,496	1,904
45 - 50	0,038	45	1,710	0,227	1,483	1,937
41 - 45	0,041	41	1,681	0,249	1,432	1,930
38 - 41	0,045	38	1,710	0,268	1,442	1,978
38-	0,049	35	1,715	0,291	1,424	2,006

STEP 3 (4.5)						
250+	0,018	250	4,500	0,108	4,392	4,608
166 - 250	0,027	166	4,482	0,163	4,319	4,645
125 - 166	0,036	125	4,500	0,216	4,284	4,716
100 - 125	0,045	100	4,500	0,270	4,230	4,770
83 - 100	0,054	83	4,482	0,325	4,157	4,807
71 - 83	0,063	71	4,473	0,380	4,093	4,853
62 - 71	0,073	62	4,526	0,435	4,091	4,961
55 - 62	0,082	55	4,510	0,491	4,019	5,001
50 - 55	0,090	50	4,500	0,540	3,960	5,040
45 - 50	0,100	45	4,500	0,600	3,900	5,100
41 - 45	0,110	41	4,510	0,659	3,851	5,169
38 - 41	0,118	38	4,484	0,711	3,773	5,195
38-	0,129	35	4,515	0,771	3,744	5,286

STEP 4 (11.2)						
250+	0,045	250	11,200	0,269	10,931	11,469
166 - 250	0,067	166	11,122	0,405	10,717	11,527
125 - 166	0,090	125	11,250	0,538	10,712	11,788
100 - 125	0,112	100	11,200	0,672	10,528	11,872
83 - 100	0,135	83	11,205	0,810	10,395	12,015
71 - 83	0,158	71	11,218	0,946	10,272	12,164
62 - 71	0,181	62	11,222	1,084	10,138	12,306
55 - 62	0,204	55	11,220	1,222	9,998	12,442
50 - 55	0,224	50	11,200	1,344	9,856	12,544
45 - 50	0,249	45	11,205	1,493	9,712	12,698
41 - 45	0,273	41	11,193	1,639	9,554	12,832
38 - 41	0,295	38	11,210	1,768	9,442	12,978
38-	0,320	35	11,200	1,920	9,280	13,120

STEP 5 (27.7)						
250+	0,111	250	27,750	0,665	27,085	28,415
166 - 250	0,167	166	27,722	1,001	26,721	28,723
125 - 166	0,222	125	27,750	1,330	26,420	29,080
100 - 125	0,277	100	27,700	1,662	26,038	29,362
83 - 100	0,334	83	27,722	2,002	25,720	29,724
71 - 83	0,390	71	27,690	2,341	25,349	30,031
62 - 71	0,447	62	27,714	2,681	25,033	30,395
55 - 62	0,504	55	27,720	3,022	24,698	30,742
50 - 55	0,554	50	27,700	3,324	24,376	31,024
45 - 50	0,616	45	27,720	3,693	24,027	31,413
41 - 45	0,676	41	27,716	4,054	23,662	31,770
38 - 41	0,729	38	27,702	4,374	23,328	32,076
38-	0,791	35	27,685	4,749	22,936	32,434

STEP 6 (67.3)						
250+	0,269	250	67,250	1,615	65,635	68,865
166 - 250	0,405	166	67,230	2,433	64,797	69,663
125 - 166	0,538	125	67,250	3,230	64,020	70,480
100 - 125	0,673	100	67,300	4,038	63,262	71,338
83 - 100	0,811	83	67,313	4,865	62,448	72,178
71 - 83	0,948	71	67,308	5,687	61,621	72,995
62 - 71	1,085	62	67,270	6,513	60,757	73,783
55 - 62	1,224	55	67,320	7,342	59,978	74,662
50 - 55	1,346	50	67,300	8,076	59,224	75,376
45 - 50	1,496	45	67,320	8,973	58,347	76,293
41 - 45	1,641	41	67,281	9,849	57,432	77,130
38 - 41	1,771	38	67,298	10,626	56,672	77,924
38-	1,923	35	67,305	11,537	55,768	78,842

STEP 7 (164)						
250+	0,656	250	164,000	3,936	160,064	167,936
166 - 250	0,988	166	164,008	5,928	158,080	169,936
125 - 166	1,312	125	164,000	7,872	156,128	171,872
100 - 125	1,640	100	164,000	9,840	154,160	173,840
83 - 100	1,976	83	164,008	11,855	152,153	175,863
71 - 83	2,310	71	164,010	13,859	150,151	177,869
62 - 71	2,645	62	163,990	15,871	148,119	179,861
55 - 62	2,982	55	164,010	17,891	146,119	181,901
50 - 55	3,280	50	164,000	19,680	144,320	183,680
45 - 50	3,644	45	163,980	21,867	142,113	185,847
41 - 45	4,000	41	164,000	24,000	140,000	188,000
38 - 41	4,316	38	164,008	25,895	138,113	189,903
38-	4,686	35	164,010	28,114	135,896	192,124

Example from Step 1:

- If stop is 50 ticks → trade 0.010 lots → risk $0.50 → commission $0.06 → net win $0.44, net loss $0.56
- If stop is 35 ticks → trade 0.014 lots → risk $0.49 → commission $0.084 → net win $0.406, net loss $0.574

The result is the same: I risk approximately $0.50 per trade at Step 1, and I earn approximately $0.44 when I win. The table ensures this consistency across all stop distances.

I run this system on 6 instruments simultaneously:
- EURUSD, GBPUSD, AUDUSD, USDCHF, NZDUSD, EURGBP

Each instrument generates 200-300 trades per week. With 6 instruments, I have 1,200-1,800 trades per week. The probability of a 7-loss streak on any single instrument is about 0.67% per sequence. With this many trades, I expect one every week or two.

But because instruments are diversified across different asset classes and currencies, losing streaks rarely happen at the same time across all instruments.

The Bottom Line:

The system is built to profit from above 50% win rate using Martingale recovery. Commission is built into every trade through a normalization table. Nano lots allow precise position sizing. Eight instruments provide diversification.

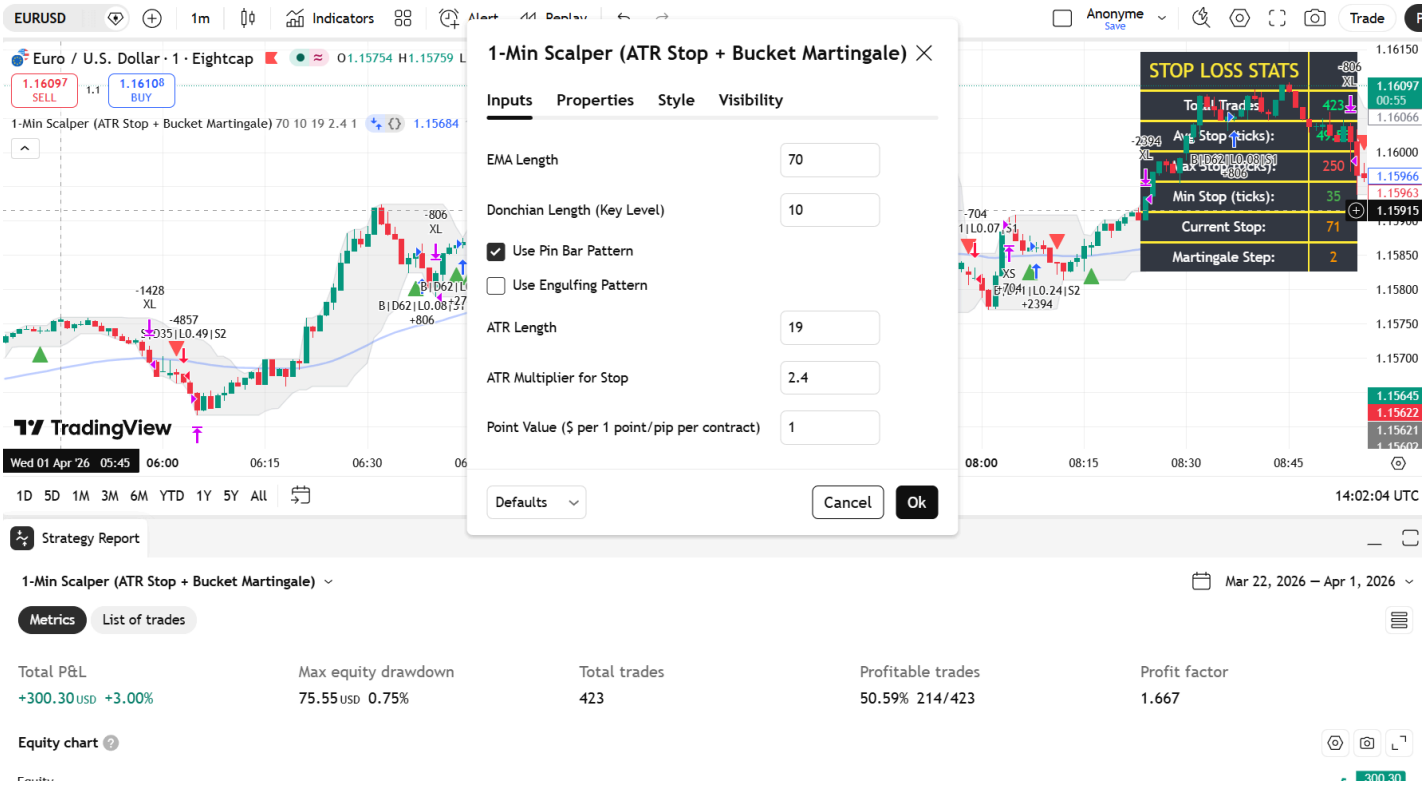
The result is a system that earns a consistent profit while keeping drawdown under 2% in normal conditions.

OPTIMIZATION FRAMEWORK - THE METRIC SYSTEM

Because market conditions change, I optimize each instrument once per week.
I built a complete optimization framework to find the best parameters for each instrument.
The process:

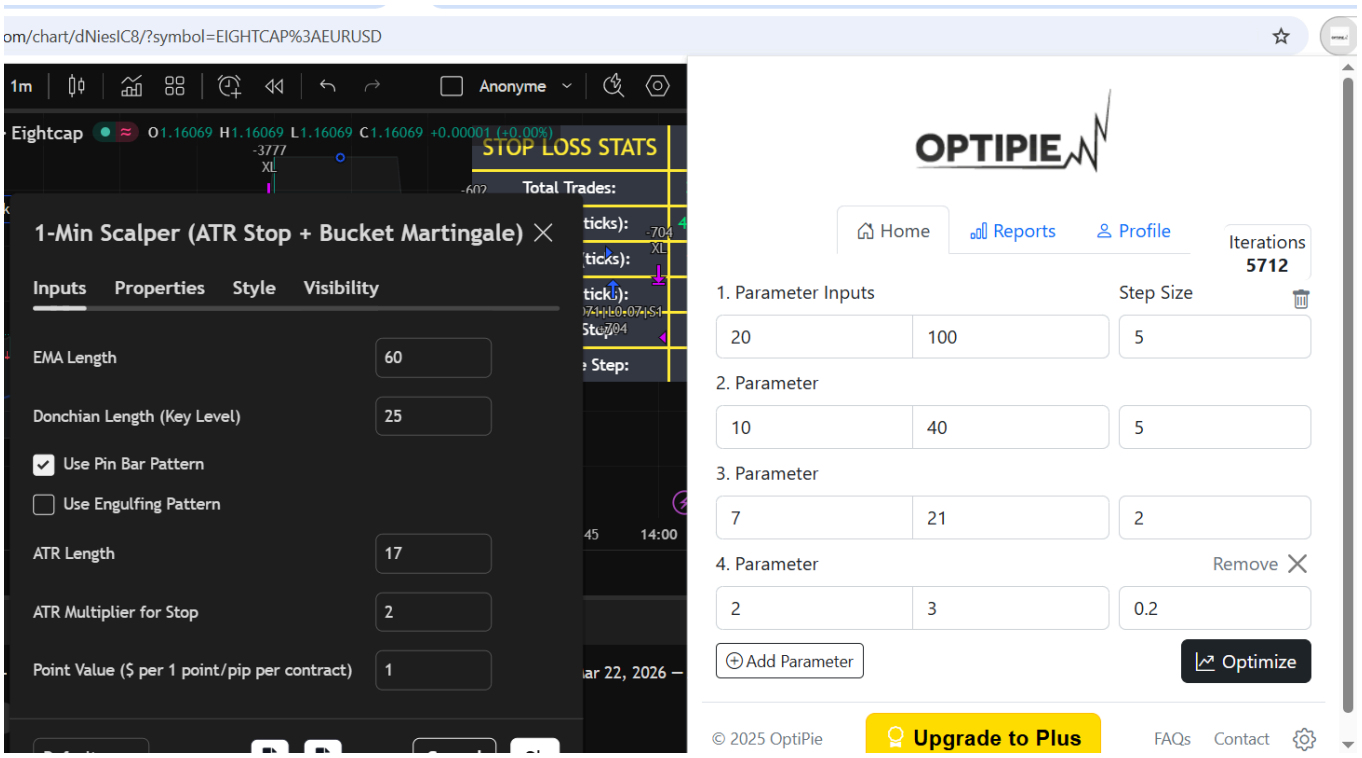
1. INPUT PARAMETERS:

- EMA Length: 20-100 (step 5)
- Donchian Length: 10-40 (step 5)
- ATR Length: 7-21 (step 2)
- ATR Multiplier: 2-3 (step 0.2)
- Pin Bar: true/false
- Engulfing: true/false



2. OPTIMIZATION EXECUTION:

I use the OptiPie Chrome extension which automates testing all 11,424 combinations per instrument. It loops through each combination, runs the backtest on TradingView, and records results.



3. SCORING ALGORITHM (Python Server):

I built a Python Flask server that receives optimization results and calculates a score for each combination:

SCORE = (Profit Factor × 30) + (Win Rate × 25) + (Drawdown Score) + (Loss Streak Penalty) + (Trades / Max Trades × 20)

Where:
- Drawdown Score = 25 × (1 - MaxDD_USD / 250)
- Loss Streak Penalty: 1-2 losses = +5 | 3 losses = 0 | 4 losses = -5 | 5 losses = -10 | 6 losses = -15 | 7+ losses = -25

This scoring system weights:
- Profitability (30%) - most important
- Win rate (25%) - Martingale needs >50%
- Drawdown (25%) - risk control
- Trade count (20%) - statistical significance
- Loss streak penalty - risk of blow-up

4. PARAMETER SELECTION:

The server sorts all combinations by score, selects the highest, and saves it to a file. I then manually update the Pine Script inputs with the winning parameters for the coming week.

PORTFOLIO - 6 INSTRUMENTS WITH CORRELATION ANALYSIS

I run the strategy on six forex instruments to diversify risk:

1. **EURUSD** - base, highest liquidity
2. **GBPUSD** - volatile, good for scalping
3. **AUDUSD** - commodity currency, Asia session
4. **USDCHF** - inverse to EURUSD, natural hedge
5. **NZDUSD** - additional commodity currency
6. **EURGBP** - cross pair, low correlation to USD

Each instrument has its own optimized parameters to account for different volatility profiles. I track correlation to ensure true diversification. The 1:1 RR applies identically across all instruments.

AUTOMATION INFRASTRUCTURE - FULL STACK

I built a complete end-to-end automation stack:

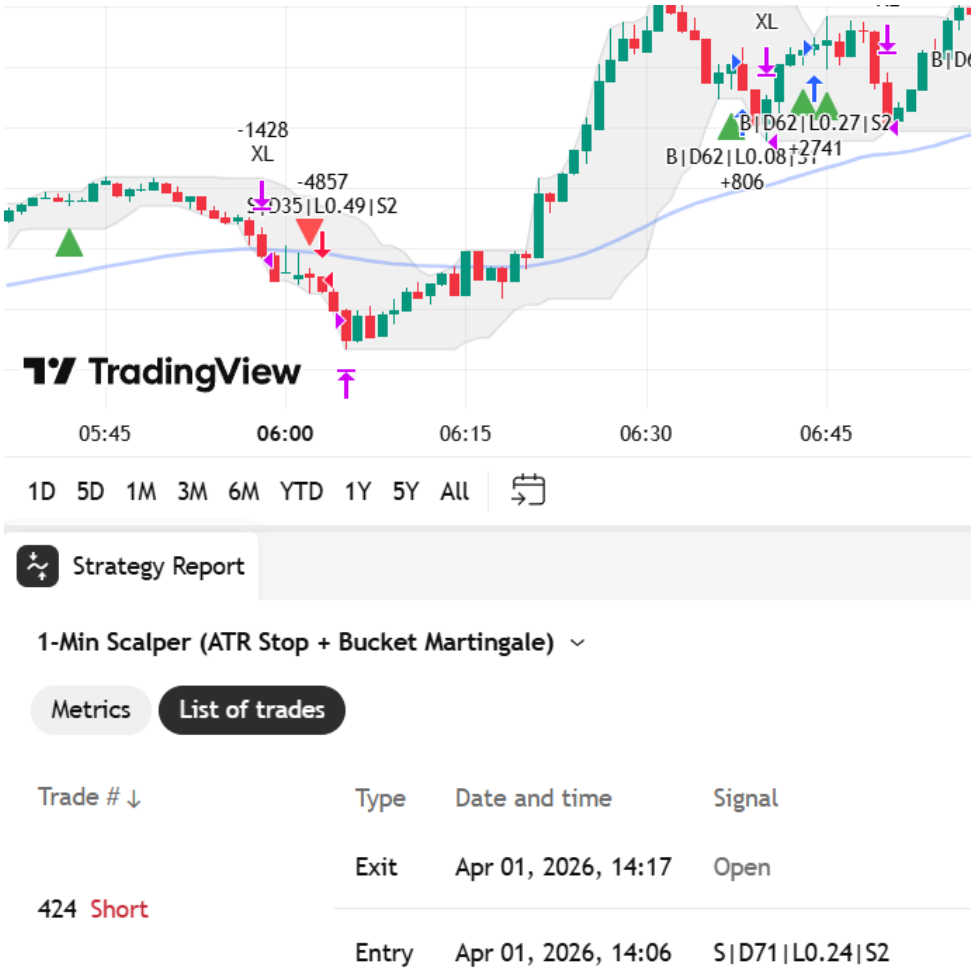
1. Pine Script Strategy:

- Outputs signals in format: "OpenB|D50|L0.001|S1"

where

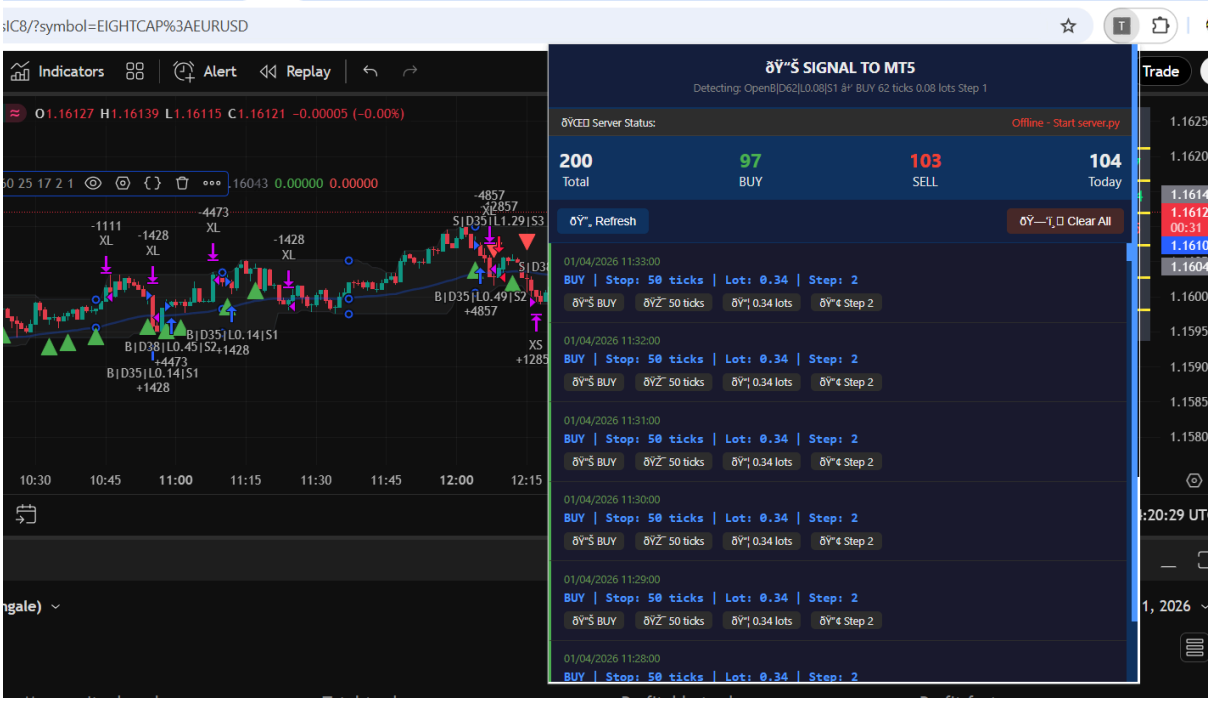
- “B” is the direction buy or sell (“B” for buy and “S” for sell)
- “D50” is the stop loss and take profit distance in ticks
- “L0.001” is the position size
- “S1” the number of step of the martingale

- Includes step number for Martingale tracking
- Fixed 1:1 risk-reward built in



2. Chrome Extension (JavaScript):

- Reads TradingView Strategy Tester table
- Detects new signals (OpenB|D...|L...|S...)
- Parses direction, stop ticks, lot size, step
- Sends to local server via HTTP POST
- Scans every 100ms for real-time detection



3. Python Flask Server:

- Receives signals at /signal endpoint
- Maintains signal queue with status tracking
- Writes to file for MT5 reading
- Handles optimization results at /optimization endpoint
- Calculates scores and saves best parameters
- Includes status endpoint for monitoring

```
127.0.0.1 - - [31/Mar/2026 17:18:40] "GET /signal HTTP/1.1" 200 -
127.0.0.1 - - [31/Mar/2026 17:18:42] "GET /signal HTTP/1.1" 200 -
127.0.0.1 - - [31/Mar/2026 17:18:43] "GET /status HTTP/1.1" 200 -
127.0.0.1 - - [31/Mar/2026 17:18:43] "GET /signal HTTP/1.1" 200 -
127.0.0.1 - - [31/Mar/2026 17:18:48] "GET /signal HTTP/1.1" 200 -
127.0.0.1 - - [31/Mar/2026 17:18:50] "GET /signal HTTP/1.1" 200 -
127.0.0.1 - - [31/Mar/2026 17:18:51] "OPTIONS /signal HTTP/1.1" 200 -

 NEW SIGNAL RECEIVED: 1774973931729|BUY|71|0.07|1
   Queue size: 1
127.0.0.1 - - [31/Mar/2026 17:18:52] "POST /signal HTTP/1.1" 200 -
 Sending signal to EA: 1774973931729|BUY|71|0.07|1
127.0.0.1 - - [31/Mar/2026 17:18:52] "GET /signal HTTP/1.1" 200 -
 Signal marked as PROCESSED: 1774973931729|BUY|71|0.07|1
127.0.0.1 - - [31/Mar/2026 17:18:52] "POST /signal/processed HTTP/1.1" 200 -
127.0.0.1 - - [31/Mar/2026 17:18:54] "GET /signal HTTP/1.1" 200 -
127.0.0.1 - - [31/Mar/2026 17:18:58] "GET /signal HTTP/1.1" 200 -
127.0.0.1 - - [31/Mar/2026 17:18:59] "GET /signal HTTP/1.1" 200 -
127.0.0.1 - - [31/Mar/2026 17:19:00] "GET /signal HTTP/1.1" 200 -
127.0.0.1 - - [31/Mar/2026 17:19:01] "OPTIONS /signal HTTP/1.1" 200 -
```

4. MT5 Expert Advisor (MQL5):

- Uses WebRequest to get signals from server
- Only processes "NEW" signals to avoid duplicates
- Parses direction, stop ticks, lot size, step
- Calculates stop loss and take profit (1:1 RR)
- Places market orders with proper position sizing
- Marks signals as "PROCESSED" after execution
- Includes position check to prevent multiple trades on same instrument

```
//+------------------------------------------------------------------+
//|                                          TradingViewBridge.mq5   |
//|                                          With Position Check     |
//+------------------------------------------------------------------+

#property copyright "TradingView Bridge"
#property version   "1.0"

#include <Trade/Trade.mqh>

// Input parameters
input int MagicNumber = 12345;
input string ServerURL = "http://127.0.0.1:8080/signal";
input string ProcessedURL = "http://127.0.0.1:8080/signal/processed";

CTrade trade;
uint lastCheckTime = 0;

//+------------------------------------------------------------------+
int OnInit() {
    trade.SetExpertMagicNumber(MagicNumber);
    Print("==========================================");
    Print("🚀 TRADINGVIEW BRIDGE v5.1 STARTED");
    Print("==========================================");
    Print("📡 Monitoring: ", ServerURL);
    Print("==========================================");
    return(INIT_SUCCEEDED);
}

//+------------------------------------------------------------------+
bool HasOpenPosition() {
    for(int i = PositionsTotal() - 1; i >= 0; i--) {
        ulong ticket = PositionGetTicket(i);
        if(PositionSelectByTicket(ticket)) {
            string symbol = PositionGetString(POSITION_SYMBOL);
```

SKILLS & TECHNOLOGIES USED

To build this system, I used:

Strategy Development

- Pine Script (TradingView) | 1-minute scalping strategy with 3-confirmation entry logic, Martingale progression, 1:1 risk-reward

Automation

- Chrome Extension (JavaScript) | Real-time signal detector reading Strategy Tester table, parsing signals, HTTP communication |

Backend Server

- Python (Flask) | Signal queue management, optimization scoring algorithm, file I/O, REST API endpoints |

Trade Execution

- MQL5 (MetaTrader) | Expert Advisor with WebRequest, position management, 1:1 stop/target calculation, duplicate prevention |

Optimization

- OptiPie Extension + Custom Python | Automated testing of 4,320 parameter combinations per instrument, scoring system with loss streak penalties |

Risk Modeling

- Excel / Custom Tables | Martingale progression tables with commission built-in, stop loss buckets, net P&L calculations |

What This Demonstrates:

- Full-stack development (frontend, backend, trading platforms)
- Quantitative reasoning (scoring systems, probability, risk models)
- Mathematical rigor (Martingale mathematics, loss streak probability, expected value)
- Resourcefulness (built everything alone with no external funding)

WHY THIS MATTERS - THE STRATEGY ARBITRAGE INSIGHT

The most important insight from this project is that with a win rate near 50% and 1:1 risk-reward, one of two strategies must be profitable:

- If win rate > 50%, Martingale on losses works
- If win rate < 50%, Martingale on wins works

I run both versions in parallel on separate PCs. The market tells me which is profitable. This eliminates the need to guess whether my strategy has edge - the math guarantees one version works.

What makes this significant is that I built this entire system with no resources, no team, no institutional backing. I did this alone, from my home, with discipline and rigorous execution. The system runs 24/7 with sub-200ms latency from signal to execution. Every component was designed, coded, and tested by me.

Now imagine what I could do with your resources.

With access to better infrastructure, more data, and a team, I am confident I can contribute real value. I understand how to build systematic frameworks, test hypotheses quantitatively, and let data drive conclusions. I have proven I can deliver results with nothing. With resources, I can scale.

This is the mindset I would bring: rigorous mathematics, disciplined execution, and the ability to build systems that work autonomously.

I am available for remote work from Tunisia and can commit even 169+ hours weekly.

Thank you for your consideration.

Iheb Mhedhbi.